

[LEDGER](#) · 2026-08-28 · INORGANIC CHEMISTRY / NANOMATERIALS

Halogen-Free Flame Retardants via Synthesis of Hexagonal Magnesium Hydroxide Nanosheets

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmarkBulk Mg(OH)₂ or ATH at 9% loading**Measured advantage**

63.5% reduction in Peak Heat Release Rate and 66.3% reduction in Smoke Production Rate at 9% loading vs minimal effect for bulk MDH.

Time to first revenue

24 months

Regulatory risk

Moderate — qualification costs reported (\$1,000,000 estimated)

Why it failed

- Voided by: Mismatched comparison

Assessment

Comparing nanosheets to bulk MDH at a 9% loading level creates an artificial baseline, as commercial bulk MDH requires 50-60% loading to function. Furthermore, UL94 certification, REACH registration, and OEM qualification cycles exceed the 18-month velocity window.

Also flagged

- Evidence rests on non-peer-reviewed sources
- no_production_runsheet

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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